Reinforcement Learning for Autonomous UAV Interception Under Target Motion Uncertainty

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***Abstract*—Autonomous counter-UAS defense requires rapid interception of hostile unmanned aerial vehicles under delayed detection, sensing noise, and stochastic target motion. This paper studies a simulated UAV-defense scenario in which a defender UAV protects a mobile ground asset from an approaching hostile UAV. The engagement is formulated as a partially observable sequential decision-making problem, and four interception strategies are evaluated under identical dynamics and terminal conditions: greedy pursuit, Kalman-enhanced greedy pursuit, Proximal Policy Optimization (PPO), and PPO with Kalman-filtered state estimates. Our results show that learning-based controllers outperform reactive pursuit baselines and improve robustness across defender speed, hostile-UAV speed, sensing radius, and measurement-noise variations. Moreover, Kalman-filtered observations are most effectively utilized when paired with a learning-based controller capable of exploiting the observations over multiple decision steps.**

***Index Terms*—Counter-UAV, Reinforcement Learning, Proximal Policy Optimization, Kalman filtering, pursuit-evasion, partial observability, autonomous interception.**

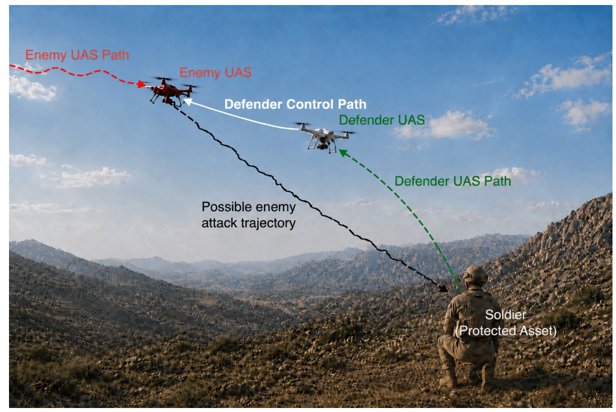

Fig. 1. Representative counter-UAS engagement considered in this work. A hostile UAV attempts to reach a protected ground asset, while a defender UAV is deployed to intercept the threat before breach.

I. Introduction

Unmanned aerial vehicles (UAVs) have become increasingly accessible due to advances in sensing, embedded computing, battery technology, and low-cost manufacturing. While UAVs support applications such as inspection, logistics, agriculture, surveillance, and emergency response, they also create security risks when operated maliciously, recklessly, or in restricted airspace. Incidents near airports, public gatherings, military installations, and critical infrastructure have increased the need for counter-UAS systems capable of detecting, tracking, and intercepting airborne threats in real time [1]–[3].

Prior work has studied UAV deployment, placement, and trajectory optimization for communication and networked aerial systems, including quantization-based trajectory design, and coverage-maximizing deployment [4]–[6]. In contrast, counter-UAS defense requires a defender UAV to react to an adversarial aerial threat under time pressure, sensing uncertainty, and safety constraints. Autonomous UAV interception is challenging because the defender must make time-critical decisions with delayed detection, noisy measurements, maneuvering target motion, and incomplete state information. Traditional counter-UAS methods often rely on rule-based pursuit, engineered tracking pipelines, or human-in-the-loop decision making, but these methods can degrade when the hostile UAV behaves unpredictably or the defender must act before complete target-state information is available [7], [8].

The problem is closely related to pursuit–evasion, target tracking, and sequential decision-making under uncertainty [9], [10]. Counter-UAS interception adds mission-specific requirements: the defender must neutralize the hostile UAV before it reaches a protected asset, avoid unsafe engagements near that asset, and operate with delayed/noisy target observations [11]. These properties make the problem naturally related to partially observable Markov decision processes (POMDPs), which are commonly used for robotic decision-making under uncertain sensing and incomplete state information [12].

State estimation is useful when target measurements are noisy. Kalman filtering and related tracking methods can reduce measurement noise and estimate target velocity when the assumed motion model is sufficiently accurate [13], [14]. However, estimation alone does not solve the control problem: even with a filtered target state, the defender must choose actions that balance speed, geometry, time-to-intercept, uncertainty, and safety.

Reinforcement learning (RL) provides a complementary approach for learning control policies through interaction with an environment. Deep RL has been applied to UAV navigation, tracking, path planning, and autonomous control [15]–[18], and recent work has explored RL-based interception of unauthorized aerial robots and dynamic UAV targets [19], [20]. However, there remains a need for controlled comparisons that isolate the separate effects of state estimation and learned sequential control.

This paper addresses that gap by developing a simulation framework for autonomous UAV defense in which a hostile UAV attempts to reach a mobile protected asset while a defender UAV seeks interception before breach. The engagement is modeled as a partially observable sequential decision-making problem with stochastic protected-asset motion, maneuvering hostile-UAV dynamics, delayed detection, noisy measurements, and safety-constrained terminal conditions. Four controllers are evaluated under identical dynamics and evaluation logic: greedy pursuit, Kalman-enhanced greedy pursuit, PPO, and PPO with Kalman-filtered state estimates.

The main contributions of this work are as follows:

- We formulate autonomous UAV interception as a partially observable sequential decision-making problem under stochastic target motion and measurement uncertainty.
- We develop a configurable counter-UAS simulation environment for evaluating defender policies under common dynamics, sensing, reward, and terminal conditions.
- We compare reactive pursuit, Kalman-enhanced pursuit, direct PPO, and PPO with Kalman-estimated observations to separate the effects of estimation and learned control.
- We evaluate robustness through parameter sweeps over defender speed, hostile-UAV speed, detection radius, and measurement-noise variance.
- We provide theoretical properties that relate the empirical sweeps to interception feasibility, sensing-radius monotonicity, and the role of Kalman belief-state information.

The remainder of this paper is organized as follows. Section II presents the system model and problem formulation. Section III describes the evaluated interception methods. Section IV states theoretical properties of the interception model. Section V details the experimental setup. Section VI reports and discusses the results. Section VII concludes the paper and outlines future research directions.

II. System Model and Problem Formulation

A. Engagement, Dynamics, and Observations

We consider a discrete-time counter-UAS engagement in a bounded two-dimensional domain $\Omega = [-L, L]^2$. The scenario contains three agents: a mobile protected asset, modeled as a soldier, a hostile UAV, and a defender UAV. At the beginning of each episode, the soldier and defender are initialized near the center of the domain, while the hostile UAV is spawned uniformly on a random boundary edge. The hostile UAV attempts to reach the soldier, and the defender attempts to intercept the hostile UAV before breach. Reflecting boundary conditions are applied so that episodes terminate because of engagement outcomes or timeout rather than map exit.

Let p_t^s, p_t^d, and p_t^e denote the positions of the soldier, defender, and hostile UAV at time step t. The simulator maintains the full latent state, including positions, maneuver bias, detection status, and estimator state. The controller, however, receives only a partially observable state representation. Before detection, enemy-related observation channels are masked. After detection, the controller receives either noisy measurements or Kalman-filtered target estimates depending on the method being evaluated.

The soldier follows an uncontrolled stochastic random-walk mobility model. This motion model is consistent with unit-level C-UAS doctrine, which treats attack avoidance, cover and concealment, movement to alternate positions, and relocation under threat as standard passive defense responses for maneuver forces facing low, slow, small UAS threats [11]. The defender is the controlled agent: each policy outputs a continuous two-dimensional heading command in $[-1, 1]^2$, which is normalized and scaled by the defender speed. This shared action interface is used by all hand-crafted and learning-based controllers.

The hostile UAV follows a stochastic pursuit model directed toward the current soldier position. Let $\hat{r}_t$ be the unit vector from the hostile UAV toward the soldier and let $\hat{r}_t^{\perp}$ be the perpendicular lateral direction. The hostile UAV maintains an autoregressive weave bias a_t, and its motion direction is

$$a_{t+1} = \rho a_t + \sigma_a \eta_t, \qquad u_t^e = \frac{\hat{r}_t + \alpha_w a_t \hat{r}_t^{\perp} + \sigma_e \omega_t}{\left\| \hat{r}_t + \alpha_w a_t \hat{r}_t^{\perp} + \sigma_e \omega_t \right\| + \varepsilon},$$

where $\eta_t \sim \mathcal{N}(0, 1)$, $\omega_t \sim \mathcal{N}(0, I_2)$, ρ controls temporal persistence, α_w controls lateral maneuvering, σ_e controls heading noise, and ε is a numerical stability constant. The hostile UAV position is then updated by moving along u_t^e at speed v_e. This model produces goal-directed but non-deterministic target motion, allowing the experiments to test robustness under maneuvering uncertainty.

Detection occurs when the defender enters the hostile UAV sensing radius, defined by $\|p_t^d - p_t^e\| \leq R_{\text{det}}$. Before detection, the hostile UAV is hidden from the controller. Once detection occurs, tracking remains active for the rest of the episode. This persistent-detection assumption isolates the effects of motion uncertainty, measurement noise, state estimation, and controller design; missed detections, track loss, clutter, and reacquisition are left for future work.

After detection, the raw target measurement is $y_t = p_t^e + \nu_t, \nu_t \sim \mathcal{N}(0, \sigma_m^2 I_2)$, where σ_m^2 is the measurement-noise variance. In the direct measurement mode, the controller observes the soldier position, defender position, detection flag, noisy target measurement, and relative defender-to-measurement vector:$o_t^{\text{raw}} = \left[p_t^s,\ p_t^d,\ b_t,\ y_t,\ y_t - p_t^d\right]$, where $b_t \in \{0, 1\}$ is the detection indicator. In the Kalman-enabled mode, the noisy measurements are processed by a constant-velocity Kalman filter, and the controller receives $o_t^{\text{KF}} = \left[p_t^s,\ p_t^d,\ b_t,\ \hat{p}_t^e,\ \hat{v}_t^e\right]$, where $\hat{p}_t^e$ and $\hat{v}_t^e$ are the estimated hostile-UAV position and velocity. All observation components are normalized before being passed to the controllers. These two observation modes allow the experiments to isolate the effect of state estimation while preserving a common control interface.

B. Objective, Reward, and Terminal Events

The defender policy is modeled as a sequential decision-maker that selects actions to maximize the expected discounted return $J(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{T} \gamma^t r_t]$, where π is the control policy,

γ is the discount factor, r_t is the scalar reward, and T is the terminal time step. The same reward and termination logic are used for all controller evaluations.

Each episode ends with one of four outcomes: safe interception, soldier capture, unsafe interception, or timeout. Define the defender–enemy distance $d_t^{de} = \|p_t^d - p_t^e\|$ and the enemy–soldier distance $d_t^{es} = \|p_t^e - p_t^s\|$. Safe interception occurs when $d_t^{de} \le R_{\text{int}}$ and $d_t^{es} > R_{\text{safe}}$. Soldier capture occurs when $d_t^{es} \le R_{\text{threat}}$. Unsafe interception occurs when the defender reaches the hostile UAV inside the protected safety region, $d_t^{de} \le R_{\text{int}}, R_{\text{threat}} < d_t^{es} \le R_{\text{safe}}$. If none of these events occurs before the maximum episode horizon $T_{\max}$, the episode terminates by timeout. In the simulator, soldier capture has highest priority, followed by unsafe interception, safe interception, and timeout. This ordering removes ambiguity when multiple conditions are simultaneously satisfied.

Terminal rewards are assigned according to the outcome:

$$r_t = \begin{cases} R_{\text{success}}, & \text{safe interception,} \\ R_{\text{caught}}, & \text{soldier capture,} \\ R_{\text{unsafe}}, & \text{unsafe interception,} \\ R_{\text{timeout}}, & \text{timeout.} \end{cases}$$

During nonterminal operation, dense shaping encourages efficient pursuit, safe geometry, and improved tracking quality when estimation is enabled:

$$r_t = \lambda_p(d_{t-1}^{de} - d_t^{de}) + \lambda_t + \lambda_k(e_{t-1}^{\text{trk}} - e_t^{\text{trk}}) + \lambda_w \left(1 - \frac{d_t^{es}}{R_{\text{warn}}}\right).$$

Here, $e_t^{\text{trk}} = \|p_t^e - \hat{p}_t^e\|$ is the Kalman tracking error. The tracking-improvement term is used only for Kalman-enabled methods after a previous tracking-error sample is available, and the proximity-warning term is applied only when $d_t^{es} < R_{\text{warn}}$. Coefficient values, engagement radii, PPO settings, and sweep configurations are reported in Section V.

III. Methods

We summarize the four controllers evaluated in the proposed counter-UAS environment. The methods are designed to separate the effects of reactive pursuit, state estimation, and learned sequential control. All controllers use the same environment dynamics, observation structure, action space, reward function, and terminal conditions defined in Section II.

A. Hand-Crafted Interception Baselines

The first baseline is a Greedy Intercept controller. Before the hostile UAV is detected, the controller uses an escort behavior and moves toward the soldier position. After detection, it moves directly toward the currently available enemy-position information. Let q_t denote the target point selected by the controller. The greedy action is the normalized direction from the defender to q_t:

$$u_t^d = \frac{q_t - p_t^d}{\|q_t - p_t^d\| + \varepsilon}.$$

For the direct Greedy baseline, $q_t = p_t^s$ before detection and $q_t = y_t$ after detection, where y_t is the noisy enemy-position measurement. This controller is computationally simple and provides a reactive pursuit reference, but it does not explicitly predict future target motion or optimize long-horizon engagement outcomes.

The second baseline is a Kalman Greedy controller. It uses the same normalized pursuit law, but after detection the target point is replaced by the Kalman-filtered enemy-position estimate. The target is selected as $q_t = p_t^s$ for $b_t = 0$ and $q_t = \hat{p}_t^e$ for $b_t = 1$. Thus, the only difference between Greedy and Kalman Greedy is the source of the post-detection target state. Comparing these two baselines isolates the effect of recursive state estimation while holding the control law fixed.

B. RL Controllers

The third controller is an RL policy trained with Proximal Policy Optimization (PPO) [16]. Unlike the hand-crafted baselines, PPO does not use an explicit escort or pursuit rule. Instead, it learns a mapping from the current observation to a continuous defender heading command through repeated interaction with the simulator. Before detection, the enemy-related observation channels are masked to zero. After detection, the direct PPO controller receives the noisy enemy-position measurement and relative defender-to-measurement geometry defined in Section II.

The fourth controller combines PPO with Kalman-based state estimation. This PPO+Kalman policy is trained and evaluated using the Kalman observation mode, where the enemy-related channels contain the estimated hostile-UAV position $\hat{p}_t^e$ and velocity $\hat{v}_t^e$ after detection. Before detection, these channels remain masked. The learned control law is therefore still a policy mapping $a_t^d \sim \pi_\theta(\cdot|o_t)$, but the observation contains filtered state information rather than raw noisy measurements.

This four-controller design enables three direct comparisons. Greedy versus Kalman Greedy measures the contribution of filtering under a fixed reactive control law. PPO versus PPO+Kalman measures the value of estimation-aware observations for a learned controller. Kalman Greedy versus PPO+Kalman measures the contribution of learned sequential decision-making when both methods use filtered target information. Detailed PPO hyperparameters, evaluation settings, and parameter-sweep configurations are reported in Section V.

IV. Theoretical Properties

In this section, we provide analytical benchmarks for the interception problem introduced in Section II. We identify geometric and information-state properties that help interpret the later experiments on defender speed, hostile-UAV speed, detection radius, and measurement noise. Full proofs are provided in Appendix A. We first provide a sufficient condition for safe post-detection interception.

Theorem 1: Consider the post-detection engagement in which the protected asset is stationary over the short interception horizon, the hostile UAV moves directly toward the protected asset at speed v_e, and the defender moves along the line of sight to the hostile UAV at speed v_d. Let d_0^{de} and

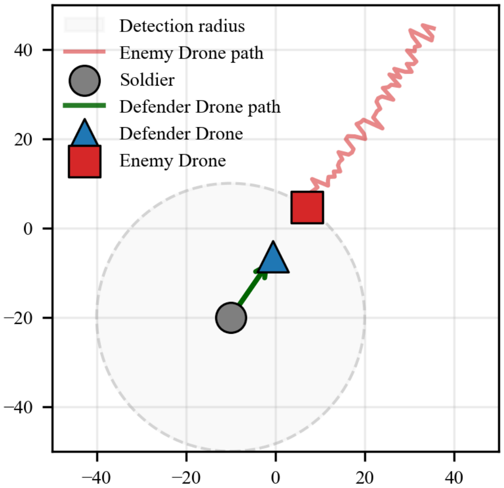

Fig. 2. Illustration of the simulated counter-UAS engagement geometry. The soldier is located near the center of the environment, the defender UAV moves outward from the soldier, and the hostile UAV approaches from outside the detection region along a noisy trajectory. The shaded circle represents the defender detection radius.

d_0^{es} denote the defender–enemy and enemy–soldier distances at detection. If

$$\frac{d_0^{de} - R_{\text{int}}}{v_d + v_e} < \frac{d_0^{es} - R_{\text{safe}}}{v_e}, \tag{1}$$

then the defender can intercept the hostile UAV before it enters the unsafe region around the protected asset. In the common case where detection occurs with the defender near the protected asset and $d_0^{de} = d_0^{es} = R_{\text{det}}$, a sufficient condition is

$$v_d > v_e \frac{R_{\text{safe}} - R_{\text{int}}}{R_{\text{det}} - R_{\text{safe}}}, \qquad R_{\text{det}} > R_{\text{safe}}. \tag{2}$$

Theorem 1 gives a baseline explanation for the speed and sensing sweeps: larger defender speed, smaller hostile-UAV speed, and larger detection radius increase the analytical safety margin. The random soldier motion, and partial observability in the simulator make the empirical problem harder than this line-of-sight benchmark, but the condition captures the same qualitative dependencies observed in the results.

A basic property of policies is monotonicity with respect to detection radii, as described via the following proposition.

Proposition 1: Let $P^\star(R_{\text{det}})$ denote the maximum achievable probability of safe interception for the same dynamics, reward, and terminal conditions when the detection radius is R_{det}. If $R_2 \geq R_1$, then $P^\star(R_2) \geq P^\star(R_1)$.

Thus, a larger sensing radius cannot reduce optimal performance because a policy can always ignore the additional early measurements and emulate a smaller-radius policy. Finite-sample training and function approximation can still produce nonmonotone empirical curves, but the optimal benchmark is monotone. We now provide an analytical motivation for the PPO+Kalman architecture.

Proposition 2: Under the post-detection linear-Gaussian tracking model used by the Kalman filter, the posterior belief $\beta_t = \mathcal{N}(\hat{x}_t, P_t)$ is a sufficient statistic for the full target measurement history. Consequently, for any history-dependent controller, there exists a belief-state controller with the same expected return that uses (p_t^s, p_t^d, β_t) instead of the complete observation history.

The Kalman filter converts a noisy measurement history into a compact target-state belief. In the implementation, the learned policy receives the posterior mean components $(\hat{p}_t^e, \hat{v}_t^e)$, which provide estimated target position and velocity to the controller. This is a compact approximation to the full belief-state policy and is especially useful when sensing noise or missed target updates make raw measurements less reliable.

V. Experimental Setup

This section describes the simulation configuration, training protocol, and evaluation procedure used to compare the four interception controllers. All methods were evaluated in the same two-dimensional UAV defense simulator using identical dynamics, sensing assumptions, reward structure, terminal conditions, and random seed ranges.

A. Simulation and Controller Configuration

Each episode begins with the soldier and defender co-located near the center of the environment and the hostile UAV initialized uniformly on a random boundary edge. The soldier follows stochastic ground motion, the hostile UAV follows the maneuvering pursuit model defined in Section II, and the defender acts according to the controller under evaluation. Unless explicitly varied in a sensitivity analysis, the nominal configuration uses a bounded square domain with half-width $L = 50$, simulation step $\Delta t = 0.5$, defender speed $v_d = 18$, hostile-UAV speed $v_e = 12$, detection radius $R_{\text{det}} = 15$, interception radius $R_{\text{int}} = 2.5$, threat radius $R_{\text{threat}} = 2.0$, and unsafe-intercept radius $R_{\text{safe}} = 3.5$. Measurement noise is modeled with nominal variance $\sigma_m^2 = 0.5$, and Kalman-enabled methods use process-noise variance $\sigma_q^2 = 1.0$ with no lead-time extrapolation.

Four controllers are evaluated: Greedy Intercept, Kalman Greedy, PPO, and PPO+Kalman. The Greedy controller uses direct reactive pursuit after detection. Kalman Greedy uses the same pursuit law but replaces the noisy target measurement with the Kalman-filtered position estimate. PPO learns a direct observation-to-action policy from raw measurement observations, while PPO+Kalman learns from observations containing Kalman-estimated target position and velocity. This design separates the effects of filtering, learned control, and estimation-aware observations.

The PPO and PPO+Kalman controllers were trained using Stable-Baselines3 PPO with an MLP policy. Both learning-based controllers used the same optimization settings, including learning rate 3×10^{-4}, discount factor $\gamma = 0.99$, rollout horizon 2048, batch size 64, PPO clipping parameter 0.2, generalized advantage estimation parameter 0.95, entropy coefficient 0.01, and value-loss coefficient 0.5. Each PPO policy was trained for 200,000 environment interaction steps. Checkpoints and deterministic validation rollouts were used during training to monitor progress, but final comparisons were performed using deterministic policy evaluation.

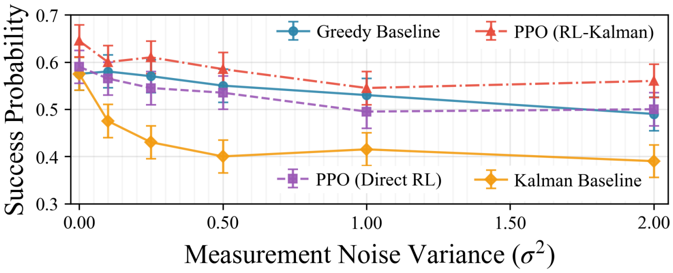

Fig. 3. Interception success rate versus measurement-noise variance. The noise sweep separates the effect of filtering alone from the effect of combining estimation with learned control.

B. Evaluation Protocol and Sensitivity Analyses

All controllers were evaluated with Monte Carlo simulation over randomized episodes. For each experiment, all methods used the same seed range, ensuring matched initial conditions and stochastic realizations. This common random numbers protocol reduces comparison variance and makes performance differences more directly attributable to controller design.

The primary metric is interception success rate, defined as the fraction of episodes ending in safe interception. Additional metrics include failure rate, timeout rate, mean episode length, detection time, intercept time, and minimum enemy–soldier distance. Kalman-enabled methods also record mean and final tracking error, and sweep experiments report standard errors and confidence intervals for probability estimates.

Robustness was evaluated using four one-dimensional sweeps and one two-dimensional feasibility study. The one-dimensional sweeps vary defender speed, hostile-UAV speed, detection radius, and measurement-noise variance, while the two-dimensional grid varies defender and hostile-UAV speeds jointly. These studies test whether learned controllers improve only nominal performance or also expand feasible engagement conditions. All experiments were implemented in Python using NumPy, Pandas, Matplotlib, and Stable-Baselines3 [21]. At inference time, all controllers remain lightweight, supporting possible future practical deployment: Baselines use vector arithmetic, Kalman methods add a four-state constant-velocity filter, and PPO methods require one feedforward MLP evaluation per decision step.

VI. Results and Discussion

This section compares the four evaluated controllers under the common Monte Carlo protocol described in Section V. The results focus on interception success, sensitivity to sensing uncertainty, and the relative contributions of state estimation and learned control.

Table I shows the nominal success-rate comparison. The direct PPO controller achieved the highest interception success rate, followed by PPO+Kalman. Both learning-based methods substantially outperformed the Greedy and Kalman Greedy baselines. In the nominal evaluation, Greedy and Kalman

TABLE I

Nominal Monte Carlo evaluation results. All controllers were evaluated over 1000 episodes; timeout rate was 0% for all methods. Confidence intervals are 95% Wilson intervals for success rate.

Controller	Success %	95% CI	Failure %	Int. time
Greedy	42.5	[39.5, 45.6]	57.5	18.4
Kalman Greedy	42.9	[39.9, 46.0]	57.1	18.1
PPO	60.3	[57.2, 63.3]	39.7	15.1
PPO+Kalman	57.7	[54.6, 60.7]	42.3	15.9

Greedy produced nearly identical success rates, indicating that filtering alone provides limited benefit when paired with a purely reactive pursuit law. By contrast, the PPO-based controllers improved both interception probability and response efficiency, reducing the mean episode length and mean intercept time relative to the hand-crafted baselines.

These results show that the dominant performance gain comes from learned sequential decision-making rather than from state estimation alone. The Greedy baseline reacts to the currently available target location, whereas PPO can learn positioning and timing behavior that improves downstream interception success. The PPO+Kalman result further indicates that filtered estimates can support learned control, but estimation-aware observations do not automatically outperform direct measurements under every nominal condition.

Fig. 3 evaluates robustness to measurement noise. As sensing noise increases, controller performance generally degrades, but the methods degrade differently. PPO+Kalman maintains strong performance across the tested noise range, suggesting that filtered position and velocity estimates are useful when the controller can exploit them over multiple decision steps. In contrast, Kalman Greedy does not consistently improve over direct Greedy. This suggests that a constant-velocity filter may introduce lag or smooth maneuver information in a way that is not well matched to instantaneous pursuit.

Additional sweeps over defender speed, hostile-UAV speed, detection radius, and the two-dimensional speed grid in Fig. 4 show a similar qualitative trend: learning-based controllers expand the feasible operating region relative to reactive pursuit. Defender-speed and enemy-speed sweeps show that the PPO controllers are especially advantageous when mobility is constrained or the hostile UAV becomes faster. Detection-radius sweep confirms that earlier target acquisition improves all methods, but sensing range alone does not close the gap between reactive and learned control.

These empirical trends are consistent with the sufficient feasibility condition in Theorem 1 and the optimal monotonicity property in Proposition 1: improved mobility and earlier detection increase the geometric margin for safe interception, although the stochastic setting and finite learned policies need not attain the idealized bound.

Overall, our simulations support three main findings. First, RL provides a substantial improvement over reactive pursuit in stochastic UAV interception. Second, Kalman filtering alone has limited value when the control policy remains greedy.

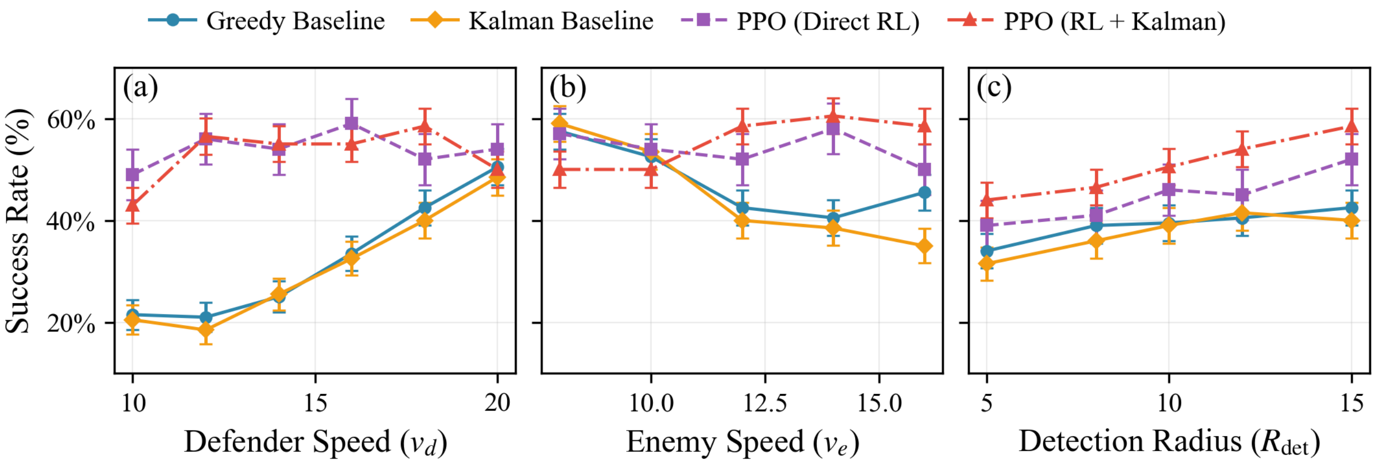

Fig. 4. Interception success rate for the four controllers under parameter sweeps of (a) defender speed v_d, (b) enemy speed v_e, and (c) detection radius r_{det}. Error bars denote 95% confidence intervals. The compared methods are Greedy Baseline, Kalman Baseline, PPO (Direct RL), and PPO (RL + Kalman).

Third, estimation is most useful when paired with a controller that can use filtered position and velocity information for sequential decision-making under uncertainty.

VII. Conclusion and Future Work

This paper studied autonomous UAV interception for protecting a mobile ground asset under delayed detection, measurement noise, and stochastic hostile-UAV motion. Using a common two-dimensional simulation environment, we compared Greedy pursuit, Kalman Greedy pursuit, PPO, and PPO with Kalman-filtered observations. The results show that PPO-based controllers improve interception performance and robustness relative to reactive pursuit baselines, and that Kalman filtering is most effective when integrated with learned sequential control. The theoretical results provide supporting benchmarks: safe interception depends on a speed–radius margin, larger detection radius improves the defender's information, and Kalman filtering gives a sufficient belief state under the linear-Gaussian tracking model. Future work will consider 3D dynamics, communication latency, multi-agent defense, adversarial policies, and real-flight validation.

References

[1] J. A. Besada, I. Campana, D. Carraminana *et al.*, "Review and simulation of counter-uas sensors for unmanned traffic management," *Sensors*, vol. 22, no. 1, p. 189, 2022.

[2] J.-E. Pierre, X. Sun, and R. Fierro, "Multi-agent partial observable safe reinforcement learning for counter uncrewed aerial systems," *IEEE Access*, vol. 11, pp. 78 192–78 206, 2023.

[3] Y. Dong, F. Wu, S. Zhang, G. Chen, Y. Hu, M. Yano, J. Sun, S. Huang, F. Liu, Q. Dai, and Z.-Q. Cheng, "Securing the skies: A comprehensive survey on anti-uav methods, benchmarking, and future directions," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2025, pp. 6725–6739.

[4] Y. Zeng, R. Zhang, and T. J. Lim, "Wireless communications with unmanned aerial vehicles: Opportunities and challenges," *IEEE Communications Magazine*, vol. 54, no. 5, pp. 36–42, May 2016.

[5] M. Mozaffari, W. Saad, M. Bennis, and M. Debbah, "Efficient deployment of multiple unmanned aerial vehicles for optimal wireless coverage," *IEEE Communications Letters*, vol. 20, no. 8, pp. 1647–1650, Aug. 2016.

[6] E. Koyuncu, M. Shabanighazikelayeh, and H. Seferoglu, "Deployment and trajectory optimization of uavs: A quantization theory approach," *IEEE Transactions on Wireless Communications*, vol. 17, no. 12, pp. 8531–8546, 2018.

[7] D. K. Nishad *et al.*, "Advanced algorithms for uav tracking of targets exhibiting start-stop and irregular motion," *Scientific Reports*, vol. 15, p. 30507, 2025.

[8] F. F. Lizzio, E. I. Trombetta, E. Capello, and Y. Fujisaki, "Comparative evaluation of multiple-model kalman filters for highly maneuvering uav tracking," *Applied Sciences*, vol. 16, no. 5, p. 2377, 2026.

[9] R. Isaacs, *Differential Games: A Mathematical Theory with Applications to Warfare and Pursuit, Control and Optimization*. New York, NY, USA: John Wiley and Sons, 1965.

[10] T. H. Chung, G. A. Hollinger, and V. Isler, "Search and pursuit-evasion in mobile robotics: A survey," *Autonomous Robots*, vol. 31, no. 4, pp. 299–316, 2011.

[11] "Counter-Unmanned Aircraft System (C-UAS)," Washington, DC, USA, Army Techniques Publication ATP 3-01.81, Aug. 2023, U.S. Army Doctrine for Countering Unmanned Aircraft Systems.

[12] M. Lauri, D. Hsu, and J. Pajarinen, "Partially observable markov decision processes in robotics: A survey," *IEEE Transactions on Robotics*, vol. 39, no. 1, pp. 21–40, 2023.

[13] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.

[14] Y. Bar-Shalom, X. R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. Wiley, 2001.

[15] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. MIT Press, 2018.

[16] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv:1707.06347*, 2017.

[17] G. Xu, W. Jiang, Z. Wang, and Y. Wang, "Autonomous obstacle avoidance and target tracking of uav based on deep reinforcement learning," *J. Intelligent Robotic Systems*, vol. 104, p. 60, 2022.

[18] S. M. Kim, J. Lee, and H. Kwon, "Tactical path planning in battlefield environments using semantic segmentation and deep reinforcement learning," *IEEE Access*, 2026.

[19] F. Giral, I. Gómez, and S. Le Clainche, "Intercepting unauthorized aerial robots in controlled airspace using reinforcement learning," p. 110985, 2025.

[20] B. Xia, I. Mantegh, and W. Xie, "Uav multi-dynamic target interception: A hybrid intelligent method using deep reinforcement learning and fuzzy logic," *Drones*, vol. 8, no. 6, p. 226, 2024.

[21] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-baselines3: Reliable reinforcement learning implementations," *J. Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.

APPENDIX A
PROOFS OF THEORETICAL RESULTS

A. Proof of Theorem 1

At detection, let the defender–enemy distance be d_0^{de}. Under the line-of-sight post-detection benchmark, the defender moves toward the hostile UAV at speed v_d, while the hostile UAV moves toward the protected asset at speed v_e. Since the defender is moving against the incoming hostile UAV, the defender–enemy separation decreases at rate $v_d + v_e$. Therefore, the first time at which the defender can enter the interception radius is

$$T_{\text{int}} = \frac{d_0^{de} - R_{\text{int}}}{v_d + v_e}. \tag{3}$$

The hostile UAV reaches the unsafe region around the protected asset when its enemy–soldier distance decreases from d_0^{es} to R_{safe}. Under the same constant-speed line-of-sight model, this time is

$$T_{\text{safe}} = \frac{d_0^{es} - R_{\text{safe}}}{v_e}. \tag{4}$$

If $T_{\text{int}} < T_{\text{safe}}$, then the defender reaches the hostile UAV while the enemy–soldier distance is still larger than R_{safe}, so the terminal event is a safe interception rather than an unsafe interception. This proves (1).

If the defender is initially near the protected asset at detection and $d_0^{de} = d_0^{es} = R_{\text{det}}$, then

$$\frac{R_{\text{det}} - R_{\text{int}}}{v_d + v_e} < \frac{R_{\text{det}} - R_{\text{safe}}}{v_e}. \tag{5}$$

Multiplying by the positive denominators and rearranging gives

$$v_e(R_{\text{safe}} - R_{\text{int}}) < v_d(R_{\text{det}} - R_{\text{safe}}), \tag{6}$$

which is equivalent to (2) when $R_{\text{det}} > R_{\text{safe}}$. This completes the proof.

B. Proof of Proposition 1

Consider two systems that differ only in detection radius, with $R_2 \geq R_1$. Couple the two systems on the same initial condition, target motion, soldier motion, process noise, and measurement noise. Whenever the hostile UAV is detected in the radius-R_1 system, it is also detected in the radius-R_2 system. The radius-R_2 system may additionally observe the hostile UAV earlier.

Take any policy feasible for the radius-R_1 system. In the radius-R_2 system, construct a policy that ignores all target observations received before the target would have entered the smaller radius R_1, and then applies the same decision rule as the radius-R_1 policy. Under the coupling, this policy reproduces the same actions and therefore the same success event distribution as the original radius-R_1 policy. Hence the best policy available under R_2 performs at least as well as this emulated policy. Taking the supremum over all policies for R_1 yields $P^\star(R_2) \geq P^\star(R_1)$.

C. Proof of Proposition 2

After detection, the Kalman tracking model has the linear-Gaussian form

$$x_{t+1} = Fx_t + q_t, \qquad q_t \sim \mathcal{N}(0, Q), \tag{7}$$

$$y_t = Hx_t + \nu_t, \qquad \nu_t \sim \mathcal{N}(0, R), \tag{8}$$

with independent Gaussian process and measurement noise. Suppose the prior belief at time t is Gaussian, $\beta_t = \mathcal{N}(\hat{x}_t, P_t)$. The Kalman prediction and update equations map $(\hat{x}_t, P_t, y_{t+1})$ to the next posterior $(\hat{x}_{t+1}, P_{t+1})$. Therefore, by induction from the initial post-detection Gaussian prior, the conditional distribution of the target state given the complete measurement history remains Gaussian and is fully described by $(\hat{x}_t, P_t)$.

The controlled process depends on the past target measurements only through the conditional distribution of the current target state. Thus, for any history-dependent policy, replacing the full measurement history by the belief b_t preserves the conditional distribution of future states, observations, rewards, and terminal events under the same action. Standard dynamic programming for belief-state POMDPs therefore admits an optimal policy that is a function of the current observable defender and protected-asset states and the target belief $(\hat{x}_t, P_t)$, rather than the entire measurement history. This proves the sufficiency claim.